

1st

```

1. import pandas as pd
2. df = pd.read_csv('Uber_Request_Data.csv')
3. df.info()
4. df.describe()
5. from sklearn import preprocessing
6. import matplotlib.pyplot as plt
7. min_max_scaler =
preprocessing.MinMaxScaler()
8. df[['Driver id', 'Request id']] =
min_max_scaler.fit_transform(df[['Driver
id', 'Request id']])
9. enc_df = pd.DataFrame()
10. enc = preprocessing.OneHotEncoder()
11. enc_arr = enc.fit_transform(df[['Pickup
point']])
12. df[['airport', 'city']] =
pd.DataFrame(enc_arr.toarray())
13. df.join(enc_df)

```

2nd

```

1. %matplotlib inline
2. import numpy as np
3. import pandas as pd
4. dic = {

    "Name":["abc","lmn","stu","xyz","efg","kl
m","rst","uvw",np.nan,"def"],
    "Maths":[96,88,np.nan,73,44,88,72,84,93,
92],
    "Reading":[77,200,88,np.nan,np.nan,91,78
,np.nan,84,93],
    "Writing":[86,np.nan,170,87,69,91,np.nan,
88,92,83],
    "Placement":[88,76,91,84,96,84,80,78,np.n
an,93],
    "Gender":["Male","Female",np.nan,"Fema
le","Female","Male","Female","Male",
"Male","Male"],
    "Joining
Date":[2018,2019,2022,2018,2017,2023,20
16,2020,2020,2019]
}
5. df = pd.DataFrame(dic)
6. df.info()
7. df["Maths"] =
df["Maths"].fillna(df["Maths"].mean())
8. df["Reading"] =
df["Reading"].fillna(df["Reading"].median(
))
9. df["Writing"] =
df["Writing"].fillna(df["Writing"].mode())
10. df["Placement"] =
df["Placement"].fillna(df["Placement"].ma
x())
11. df.dropna(inplace = True)
12. df.boxplot(["Maths","Reading","Writing",
"Placement"])
13. from sklearn.preprocessing import
OneHotEncoder
14. enc = OneHotEncoder()
15. enc_arr = enc.fit_transform(df[['Gender']])
16. enc_pd = pd.DataFrame(enc_arr.toarray())
17. df = df.join(enc_pd)
18. from sklearn.preprocessing import
MinMaxScaler
19. scaler = MinMaxScaler()
20. df[['Maths','Reading','Writing','Placement']] =
scaler.fit_transform(df[['Maths','Reading','
Writing','Placement']])

```

3rd

```

1. import pandas as pd
2. from sklearn.preprocessing import
OneHotEncoder
3. df = pd.read_csv('Mall_Customers.csv')
4. df.groupby(['Gender'])['Age'].mean()

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5. df.groupby(['Gender'])['Age'].median()
6. df.groupby(['Gender'])['Age'].std()
7. df.groupby(['Gender'])['Age'].min()
8. df.groupby(['Gender'])['Age'].max()
9. df.groupby(['Gender'])['Annual Income
(k$)'].mean() median std min max
10. df.groupby(['Gender'])['Spending Score (1-
100)'].mean() median std min max
11. df.groupby(['Gender']).mean() median std
min max
12. enc = OneHotEncoder()
13. enc_df =
pd.DataFrame(enc.fit_transform(df[['Gend
er']]).toarray())
14. df1 = df.join(enc_df)
15. iris_df = pd.read_csv('iris.csv')
16. iris_df.groupby(['variety']).mean() median
std min max
17. iris_df[iris_df.variety=='Setosa'].describe()
18. iris_df[iris_df.variety=='Versicolor'].describ
e()
19. iris_df[iris_df.variety==
'Virginica'].describe()

```

4th

```

1. import pandas as pd
2. from sklearn.preprocessing import
MinMaxScaler
3. from sklearn.linear_model import
LinearRegression
4. df = pd.read_csv('BostonHousing.csv')
5. info describe isnull.sum
6. scaled_df = pd.DataFrame()
7. scaler = MinMaxScaler()
8. scaled_df[['crim', 'zn', 'indus', 'chas', 'nox',
'rm', 'age', 'dis', 'rad', 'tax',
'ptratio', 'b', 'lstat',]] =
pd.DataFrame(scaler.fit_transform(df[['cri
m', 'zn', 'indus', 'chas', 'nox', 'rm', 'age',
'dis', 'rad', 'tax',
'ptratio', 'b', 'lstat',]]))
10. x = scaled_df
11. y = df['medv']
12. from sklearn.model_selection import
train_test_split
14. x_train, x_test, y_train, y_test =
train_test_split(x,y,test_size=0.2,random_s
tate=42)
15. model = LinearRegression()
16. model.fit(x_train.values,y_train.values)
17. y_train_preds =
model.predict(x_train.values)
18. y_test_preds =
model.predict(x_test.values)
19. df1 = pd.DataFrame({'Actual' :
y_train,'Predicted' : y_train_preds})
20. df2 = pd. DataFrame({'Actual' :
y_test,'Predicted' : y_test_preds})
21. from sklearn.metrics import
mean_squared_error, r2_score
22. train_accuracy_mse = 100-
mean_squared_error(y_train,y_train_pred
s)
23. test_accuracy_mse = 100-
mean_squared_error(y_test,y_test_preds)
24. train_accuracy_r2 = 100-
mean_squared_error(y_train,y_train_pred
s)
25. test_accuracy_r2 = 100-
mean_squared_error(y_test,y_test_preds)
26. import matplotlib.pyplot as plt
27. plt.scatter(y_train,y_train_preds,c='red',m
arker='o',label='Train Data')
28. plt.scatter(y_test,y_test_preds,c='blue',ma
rker='s',label='Test Data')
29. plt.plot()
30. plt.show()

```

5th

```

1. import pandas as pd from
sklearn.model_selection import
train_test_split
2. from sklearn.preprocessing import
StandardScaler
3. from sklearn.linear_model import
LogisticRegression
4. from sklearn.metrics import
confusion_matrix,accuracy_score
5. df=pd.read_csv(r"C:\Users\Dell\Social_Net
work_Ads.csv")
6. x=df.drop(['Purchased'],axis=1)
7. y=df['Purchased']
8. x_train,x_test,y_train,y_test=train_test_spl
it(x,y,test_size=0.25)
9. sc=StandardScaler()
10. x_train=sc.fit_transform(x_train)
11. x_test=sc.transform(x_test)
12. classifier=LogisticRegression()
13. classifier.fit(x_train,y_train)
14. y_test_pred=classifier.predict(x_test)
15. classifier.predict([[ -0.92863358,
0.31770514]])
16. classifier.predict([[2.01088048,
0.22907072]])
17. matrix=confusion_matrix(y_test,y_test_pr
ed)
18. score=accuracy_score(y_test,y_test_pred)
19. from sklearn.metrics import
classification_report
20. print(classification_report(y_test,y_test_pr
ed))
21. print('True Positive:', matrix[0][0])
22. print('True Negative:', matrix[1][1])
23. print('False Positive:', matrix[0][1])
24. print('False Negative:', matrix[1][0])
25. print('Accuracy:', score) print('Error
Rate:', 1-score) precision =
matrix[0][0]/(matrix[0][0]+matrix[0][1])
print('Precision:', precision)

```

6th

```

1. import pandas as pd
2. from sklearn.model_selection import
train_test_split
3. from sklearn.metrics import
confusion_matrix,classification_report,acc
uracy_score
4. from sklearn.preprocessing import
LabelEncoder
5. from sklearn.naive_bayes import
GaussianNB
6. df=pd.read_csv(r"C:\Users\Dell\Iris.csv")
7. label_encoder=LabelEncoder()
8. df['Species']=label_encoder.fit_transform(
df['Species'])
9. x=df.drop('Species',axis=1)
10. y=df.Species
11. x_train,x_test,y_train,y_test=train_test_spl
it(x,y,test_size=0.2)
12. gaussian=GaussianNB()
13. gaussian.fit(x_train,y_train)
14. y_test_pred=gaussian.predict(x_test)
15. gaussian.predict([[123,7.7,2.8,6.7,2.0]])
16. matrix=confusion_matrix(y_test,y_test_pr
ed)
17. accuracy=accuracy_score(y_test,y_test_pre
d)
18. error=1-accuracy
19. print(classification_report(y_test,y_test_pr
ed))

```

7th

```

1. !pip install nltk
2. import nltk
3. from nltk.tokenize import sent_tokenize
4. from nltk.tokenize import word_tokenize
5. from nltk.corpus import stopwords
6. from nltk.stem import PorterStemmer

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7. from nltk.stem import
   WordNetLemmatizer
8. import re
9. from sklearn.feature_extraction.text
   import TfidfVectorizer
10. nltk.download('punkt')
11. nltk.download('stopwords')
12. nltk.download('wordnet')
13. nltk.download('averaged_perceptron_tagg
   er')
14. text="Tokenization replaces a sensitive
   data element, for example, a bank
   account"
15. nltk.download('punkt_tab')
16. sentences=sent_tokenize(text)
17. print(sentences)
18. words=word_tokenize(text)
19. print(words)
20. stop_words=set(stopwords.words('english'
   ))
21. print(stop_words)
22. text="How to remove stopwords with NLTK
   library in PYTHON?"
23. text=re.sub('[^a-zA-Z]', '',text)
24. tokens=word_tokenize(text.lower())
25. filtered_text=[]
26. for w in tokens:
27.     if w not in stop_words:
28.         filtered_text.append(w)
29. print('tokenized text:',tokens)
30. print('filtered text:',filtered_text)
31. nltk.download('averaged_perceptron_tagg
   er_eng')
32. words1=['ram','before','eating','waiting']
33. for w in words1:
34.     print(w,":",nltk.pos_tag([w]))
35. ps=PorterStemmer()
36. for w in words1:
37.     print(w,":",ps.stem(w))
38. lemmatizer=WordNetLemmatizer()
39. for w in words1:
40.     print(w,":",lemmatizer.lemmatize(w,pos='v'
   ))
41. paragraph="India is a vast country with
   second highest populati on in the world. It
   is a country with diverse cultures,
   traditions and beliefs. People in India cel
   ebrate unity in diversity. Festivals like
   Diwali, Holi, Navratri, Ramzan, Christmas
   etc. are celebrated by people across India
   and create a sense of brotherhood and
   cultural unity. Each festiva l has its religious
   and cultural importance."
42. wn=WordNetLemmatizer()
43. sentences=sent_tokenize(paragraph)
44. cleaned_para=[]
45. for i in range (len(sentences)):
46.     review=re.sub('[^a-zA-Z]', '' ,sentences[i])
47.     review=review.lower()
48.     review=review.split()
49.     review = [wn.lemmatize(word) for word in
   review if not word in set(stopwords
   review=' '.join(review)
   cleaned_para.append(review)
50. print(cleaned_para)
51. tfidf=TfidfVectorizer()
52. x=tfidf.fit_transform(cleaned_para).toarray
   ()
53. print(x)

```

8th

```

1. import pandas as pd
2. import numpy as np
3. import matplotlib.pyplot as plt
4. import seaborn as sns
5. titanic=sns.load_dataset('titanic')
6. sns.displot(titanic['age'],bins=10)
7. sns.displot(titanic['fare'],bins=10)
8. sns.countplot(x='survived',data=titanic)

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9. sns.countplot(x='sex',data=titanic)
10. sns.countplot(x='pclass',data=titanic)

```

9th

```

1. import pandas as pd
2. import numpy as np
3. import matplotlib.pyplot as plt
4. import seaborn as sns
5. titanic=sns.load_dataset('titanic')
6. titanic.head()
7. sns.boxplot(x='sex',y='age',data=titanic)
8. sns.boxplot(x='sex',y='age',data=titanic,hu
   e='survived')

```

10th

```

1. import pandas as pd
2. import numpy as np
3. import matplotlib.pyplot as plt
4. import seaborn as sns
5. iris=sns.load_dataset('iris')
6. iris.head()
7. sns.histplot(iris['sepal_length'],bins=10)
8. plt.show()
9. sns.histplot(iris['sepal_width'],bins=10)
10. plt.show()
11. sns.histplot(iris['petal_length'],bins=10)
12. plt.show()
13. sns.histplot(iris['petal_width'],bins=10)
14. plt.show()
15. sns.boxplot(x='species',y='sepal_length',da
   ta=iris)
16. plt.show()
17. sns.boxplot(x='species',y='sepal_width',dat
   a=iris)
18. plt.show()
19. sns.boxplot(x='species',y='petal_length',da
   ta=iris)
20. plt.show()
21. sns.boxplot(x='species',y='petal_width',dat
   a=iris)
22. plt.show()

```